



@dandjlab

THE FREE STACK

- 01 **ffmpeg** clean the audio, then find the pauses `mono 16khz wav / silencedetect`
- 02 **whisper** the words and the timing `faster-whisper 1.2.1 / large-v3 pinned`
- 03 **pyannote** who speaks when `pyannote.audio 4.0.7 / community-1`
- 04 **numpy** pause, energy, pitch and pacing for every word `measured inside the frozen boundaries`
- 05 **modal** one rented gpu runs it all `warm worker / four clips in parallel`

THE TEST

One hour and 44 minutes of the same audio, transcribed six times. Three times with the custom free stack on a rented gpu. Three times through the Gemini 3.5 Transcribe api, one request per clip.

WHAT CAME BACK

- 01 **The Free Stack cost 39% less (free with Modal's \$30 free credits).**

Free Stack		\$0.319
Gemini 3.5 Transcribe		\$0.520
But		

- 02 **The Free Stack gave the same answer every time. Gemini did not.**

Three cold runs of the Free Stack returned byte-identical words, timings and speaker files. Gemini's word count changed between repeats, and its returns carried hundreds of zero-duration words and dozens of timestamps out of order.

- 03 **One request said completed and was missing nine minutes of speech.**

Nine minutes and 34 seconds out of the middle of one clip, gone, with no error returned. Same boundary in all three runs of that mode. The Free Stack found 1,595 words in that same stretch.

FULL HEAD TO HEAD

Same 103.97 minutes of real footage. Three repeat trials per configuration.

	free stack	gemini 3.5 transcribe
median processing time	5m 57s	4m 06s to 4m 12s
estimated cost per run	\$0.319	\$0.520
cost difference	38.7% cheaper	—
monthly cost to you	\$0 within modal's \$30 credit	paid api
word output	17,494 every trial	16,043 to 17,979
repeatability	words and speakers byte-identical	changed between trials
zero-duration words	29	516 to 649
timeline reversals	0	29 to 72
overlaps	0	46 to 88
missing section	none found	9m 34.4s skipped, 3 of 3 times
diarization	speaker identities across the whole set	per clip
tone and pacing	included as prosody	not provided
transcription api calls	zero	files + interactions apis

Gemini was about two minutes faster. Its words-only mode skipped the same 9m 34.4s section on all three runs while still reporting completed. On that clip it returned 750 to 767 words, against 2,394 from the Free Stack.

HAND THIS TO CLAUDE OR CODEX

Build the Free Stack to this spec. Use the exact package versions, model revisions and inference settings on this page. Do not substitute anything on the never list. When you are done, run the same audio three times and show me the checksums.

PINNED RUNTIME

environment	Debian / Python 3.11	cuda 12 / cudnn 9
audio	ffmpeg	mono pcm16 wav / 16 khz
pauses	ffmpeg silencedetect	noise=-35dB / d=0.25 / written beside words.json, never into it
package	faster-whisper	1.2.1
backend	CTranslate2	4.8.1
model	faster-whisper-large-v3	rev 53ecf83a5bedc5597eb8c8b34eac29e5345520ff
precision	compute_type=float32	not fp16, not int8
execution	WhisperModel(num_workers=4)	native transcribe, four clips at once
inference	vad_filter=False	word_timestamps=True
	temperature=0.0	condition_on_previous_text=False
speakers	pyannote.audio	4.0.7
model	speaker-diarization-community-1	rev 3533c8cf8e369892e6b79ff1bf80f7b0286a54ee
batching	segmentation=8 / embedding=8	one pass over the whole batch
numeric	torch 2.8.0 / numpy 2.4.6	torchaudio 2.8.0 / soundfile
access	hugging face token	community-1 weights are gated
compute	one warm NVIDIA L40S	16 cpu / 64 gib / models load once per worker

ORDER OF OPERATIONS

- | | | |
|----|-------------------|---|
| 01 | normalize | ffmpeg to mono 16 khz. the original file is never rewritten |
| 02 | transcribe | one model in memory, four clips at once, settings above |
| 03 | freeze | write word, start, end to disk and never edit them again |
| 04 | pauses | ffmpeg silencedetect over the same audio, written to its own file. see page 5 |
| 05 | assemble | every clip into one lane, 2 s of silence between, keep the offset map |
| 06 | diarize | one speaker pass over that lane, then map turns back to clip time |
| 07 | assign | each word takes the speaker with the greatest positive overlap |
| 08 | measure | one prosody row per word: pause, duration, energy, voicing, pitch and slope |
| 09 | confirm | a human names the speakers |
| 10 | publish | hash the output, then write the transcript |

THE PAUSES

VAD off keeps every word. The side effect: Whisper glues each pause onto a neighbouring word, so the timestamps carry no silence. A cut that starts on one of those words plays dead air before anyone speaks.

MEASURED, ONE 17.6 MINUTE CLIP, LAV MIC, QUIET ROOM

silences of 0.4 s or longer in the audio	289
marked as a real gap between words	124
glued to the start of the next word, so that word is stamped early. median 0.6 s, worst 3.9 s	76
glued to the end of the previous word, so that word ends late	70
unclear or mid-word	19
one representative minute: audible pauses in the audio, gaps in the timestamps across 198 words	17 and 0

THE FIX: MEASURE, DO NOT FILTER

Do not rewrite Whisper's word times. Run one more pass over the same audio and write the silences beside the words as their own file. One command, about a second per clip:

```
ffmpeg -i clip.mov -vn -af "silencedetect=noise=-35dB:d=0.25" -f null -
```

```
{"method": "ffmpeg-silencedetect-v1", "noise_db": -35.0, "min_silence_s": 0.25,
"silences": [[541.29, 542.22], [543.07, 543.35], [550.25, 551.91]]}
```

A floor of -35 dB and a minimum of 0.25 s caught every audible pause in a quiet room. A noisy location needs a higher floor, so keep both as settings and write the values used into the file. An energy threshold over decoded audio is bit-identical across workers. A speech detector is a classifier, which is the same reason VAD is off.

WHAT THE CUT DOES WITH IT

If a cut's in-point lands inside a silence, move it to the end of that silence. If the out-point lands inside one, move it to the start. Cap the move at 4 s, record the delta, and never touch the word indices. On ten clips: 2,458 silences found in 7.4 s, scanned twice with identical output, 41 of 64 cut-ins moved by a median 0.59 s, and 26 of 28 cut-ins now start on a real pause.

IF THE BUILD IS RIGHT, THESE ARE TRUE

- 01 the same audio, run three times, gives byte-identical word and speaker files
- 02 every word starts before it ends, and no start runs backward
- 03 no two words overlap in time
- 04 the word count does not change between runs
- 05 a person listens to thirty seconds and confirms the transcript matches

NEVER

- VAD on.** a vad path omitted real speech, so production runs without it
- 30-second chunking.** it lost words at the boundary, so whole clips are transcribed
- FP16.** it produced timestamp and word-sequence variance across fresh workers
- Speaker IDs per clip.** speaker 1 in clip 3 is not speaker 1 in clip 7
- A model naming people.** diarization hears voices, not identities
- Fuzzy text to time.** use the timestamps you froze
- Editing word times to add pauses.** measure the silence in its own file and snap to it. the words stay raw